GATE
CS & DA

Artificial Intelligence

Miscellaneous

Lecture No. 1

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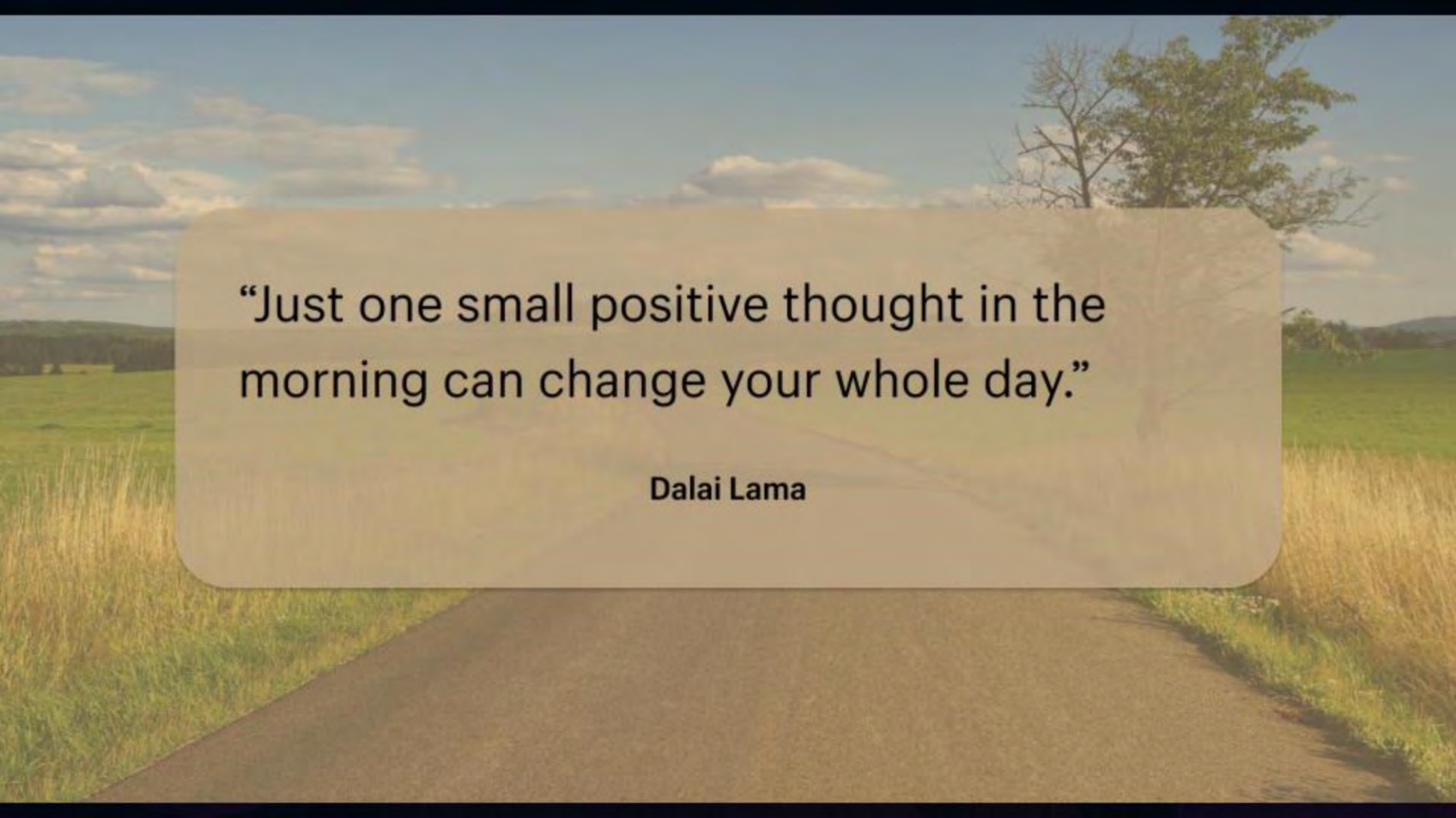
Recap of Previous Lecture

- Topic logic
- Topic
- Topic
- Topic
- Topic

Turn on Slide map

Topics to be Covered

- Topic Bayesian N/W
- Topic
- Topic
- Topic
- Topic


“Just one small positive thought in the morning can change your whole day.”
Dalai Lama

Q.54	Let *game(ball, rugby)* be true if the ball is used in rugby and false otherwise. Let *shape(ball, round)* be true if the ball is round and false otherwise. Consider the following logical sentences: $s1$: $\forall ball\ \neg game(ball, rugby) \Rightarrow shape(ball, round)$ $s2$: $\forall ball\ \neg shape(ball, round) \Rightarrow game(ball, rugby)$ $s3$: $\forall ball\ game(ball, rugby) \Rightarrow \neg shape(ball, round)$ $s4$: $\forall ball\ shape(ball, round) \Rightarrow \neg game(ball, rugby)$ Which of the following choices is/are logical representations of the assertion, "All balls are round except balls used in rugby"?
(A)	$s1 \wedge s3$
(B)	$s1 \wedge s2$
(C)	$s2 \wedge s3$
(D)	$s3 \wedge s4$

$(S_1 \equiv S_2)$
$(S_3 \equiv S_4)$

$(S_1 \text{ and } S_3)$

(S_1, S_3) (S_2, S_4)
(S_2, S_3) (S_1, S_4)

Some rules of Logic

Equivalence	Name
$p \wedge T \equiv p, \quad p \vee F \equiv p$	Identity laws
$p \vee T \equiv T, \quad p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p, \quad p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \vee q) \equiv \neg p \wedge \neg q$ $\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$	Negation laws

Some rules of Logic

Logical Equivalences Involving Conditional Statements

$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg p \rightarrow q$
$p \wedge q \equiv \neg(p \rightarrow \neg q)$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Logical Equivalences Involving Biconditional Statements

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Some rules of Logic

Steps in Conversion of Propositions to Conjunctive Normal Form

- Eliminate biconditionals
  - Turn $(\alpha \leftrightarrow \beta)$ into $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$.
- Eliminate implications
  - Turn $(\alpha \rightarrow \beta)$ into $\neg\alpha \vee \beta$.
- Move negation inwards until only literals are being negated (and not clauses), using De Morgan's Laws.
  - Turn $\neg(\alpha \wedge \beta)$ into $\neg\alpha \vee \neg\beta$

Probabilistic Reasoning in Artificial Intelligence

Probabilistic reasoning in Artificial Intelligence (AI) refers to the use of probability theory to model and manage uncertainty in decision-making processes. This approach is fundamental in creating intelligent systems that can operate effectively in complex, real-world environments where information is often incomplete or noisy.

Probabilistic Reasoning in Artificial Intelligence

Definition of Uncertainty in AI, uncertainty arises when the information needed to make a decision is incomplete, noisy, or ambiguous. It can originate from:

- Imperfect Information: Data might be incomplete or unreliable.
- Ambiguity: Several possible interpretations of the data.

Probabilistic Reasoning in Artificial Intelligence

- Probabilistic reasoning provides a mathematical framework for representing and manipulating uncertainty.
- Unlike deterministic systems, which operate under the assumption of complete and exact information, probabilistic systems acknowledge that the real world is fraught with uncertainties.
- By employing probabilities, AI systems can make informed decisions even in the face of ambiguity.

Probabilistic Reasoning in Artificial Intelligence

- Reasoning under uncertainty is a fundamental concept in artificial intelligence (AI) that involves making decisions or drawing conclusions when the information available is incomplete, ambiguous, or subject to change.
- Unlike deterministic systems, where outcomes are predictable and certain, AI systems often operate in environments where uncertainty is a significant factor.

Probabilistic Reasoning in Artificial Intelligence

- We have many methods to be used in the probabilistic reasoning
- Bayesian Network
- Markov chain

Probability

* $P(A, B)$ = Joint Prob.
* $P(A/B) = \dfrac{P(A, B)}{P(B)}$

Bayesian Networks

- A Bayesian Network (BN), also called a Belief Network, is a graphical model that represents probabilistic relationships among a set of variables using a directed acyclic graph (DAG).
- The nodes in the graph represent random variables, and the edges represent conditional dependencies between these variables.
- Each node in a Bayesian Network has a Conditional Probability Distribution (CPD), which defines the probability of the variable given its parents in the network.

Bayesian Networks

donot depent on any one

dependent on cloudy

DAG ⇒

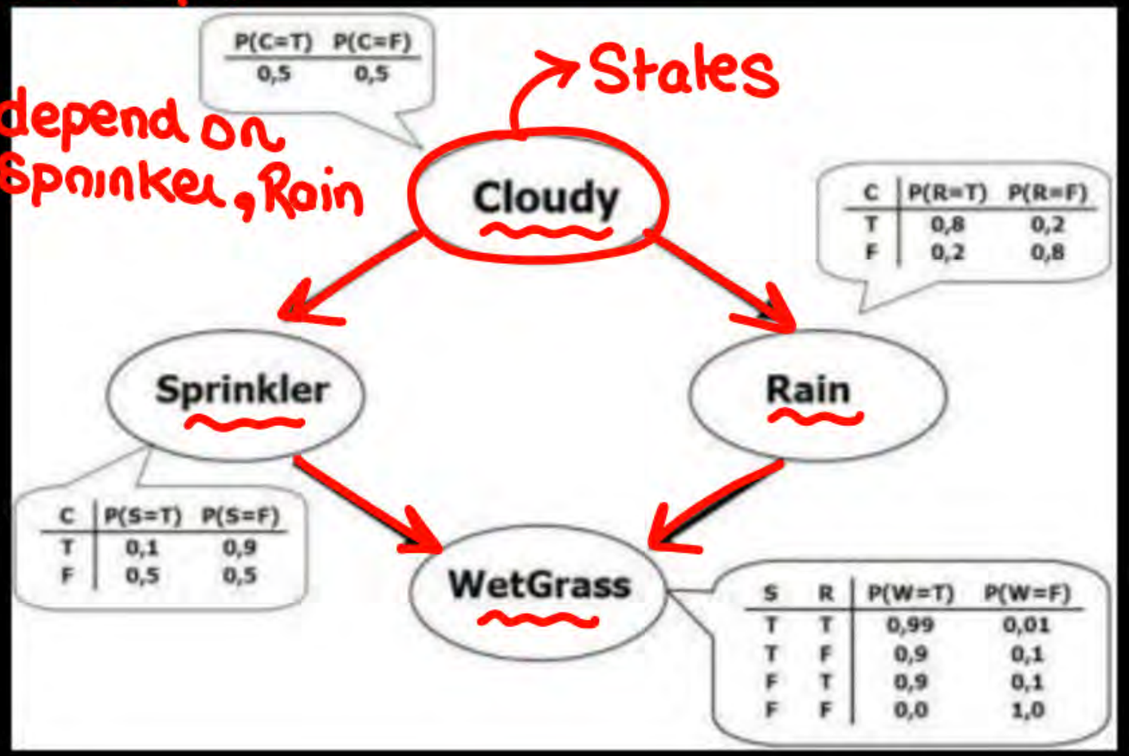
- Example
- Cloudy, Sprinkler, Wet Grass, Rain are the Random variable.
- The arrow show the dependencies.

Cloudy

depend on Sprinkler, Rain

⇒ nodes : RV

⇒ arrow : dependence



Bayesian Networks

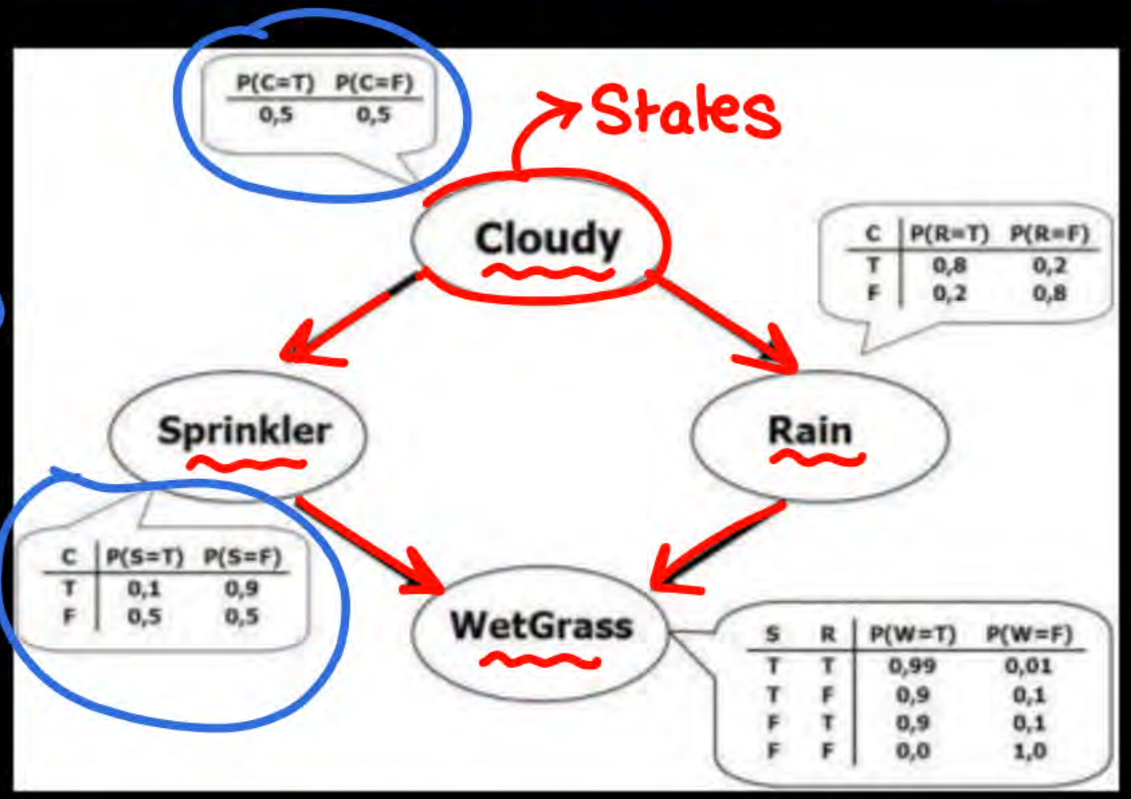
- Table written with all RV's ⇒

S R	P(W=T)
FF	.99
FT	.9
TF	.9
TT	.9

$P(W=T/S=F,R=F)$

C	P(S=T)	P(S=F)
T	.1	.9
F	.5	.5

$P(S=T/C=T)$ $P(S=T/C=F)$



What is ⇒

$$\Rightarrow P(A,B) = P(A=T, B=T)$$

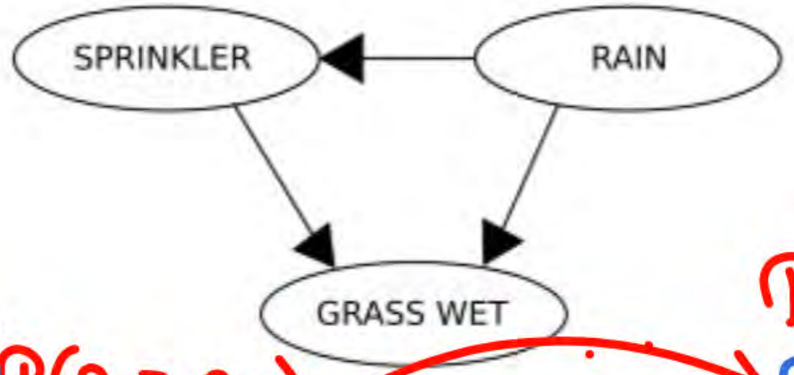
$$= P(A=T/B=T) \cdot P(B=T)$$

- $P(R|G)$
- $P(R=T/G=T)$

Bayesian Networks

- P(G,S,R)
- P(R/G)

RAIN	SPRINKLER T	SPRINKLER F
F	0.4	0.6
T	0.01	0.99



RAIN T	RAIN F
0.2	0.8

SPRINKLER	RAIN	GRASS WET T	GRASS WET F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

• $P(G,S,R)$

$P(G=T, S=T, R=T) = \boxed{P(G=T/S=T,R=T)}\, P(S=T, R=T)$

$= P(G=T/S=T,R=T)\, P(S=T/R=T).\, P(R=T)$

$= .99 \times .01 \times .2$

$= 1.98 \times 10^{-3}$

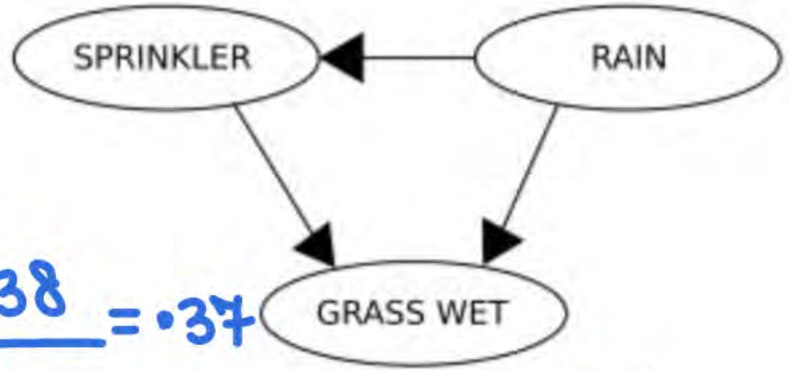
$P(S=T, R=T)$

$P(S=T/R=T).P(R=T)$

Bayesian Networks

- P(G,S,R)
- P(R/G)

RAIN	SPRINKLER T	SPRINKLER F
F	0.4	0.6
T	0.01	0.99



RAIN T	RAIN F
0.2	0.8

SPRINKLER	RAIN	GRASS WET T	GRASS WET F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

$$P(R/G) \Rightarrow \frac{P(R=T, G=T)}{P(G=T)} = \frac{.16038}{.43618} = .37$$

$P(R=T, G=T) \rightarrow$

$$\boxed{P(R=T, G=T, S=T) + P(R=T, G=T, S=F)}$$

$$\boxed{P(A) = P(A/B=0)P(B=0) + P(A/B=1)P(B=1)}$$

$$\Rightarrow \overset{.99}{P(G=T/R=T, S=T)} \cdot P(R=T, S=T) \rightarrow \overset{.01 \times .2}{P(S=T/R=T)P(R=T)}$$

$$+ \overset{.8}{P(G=T/R=T, S=F)} \cdot P(R=T, S=F) \rightarrow \overset{.99 \times .2}{P(S=F/R=T)P(R=T)}$$

$$\Rightarrow .16038$$

$$P(G=T) = \overset{0}{P(G=T/R=F,S=F)}\,\underline{P(R=F,S=F)}$$

$$+ \overset{.8}{P(G=T/R=F,S=T)}\,\underline{P(R=F,S=T)}$$

$$+ \overset{.9}{P(G=T/R=T,S=F)}\,P(R=T,S=F)$$

$$+ \overset{.99}{P(G=T/R=T,S=T)}\,P(R=T,S=T)$$

$$\Rightarrow \boxed{(.43618)}$$

$$P(S=T/R=F)P(R=F) \quad .4 \times .8$$

$$P(S=F/R=T)P(R=T) \quad .99 \times .2$$

$$P(S=T/R=T)P(R=T) \quad .01 \times .2$$

Recall

8 Cases $\Leftarrow$ $\underbrace{A, B, C}_{T/F}$

$$P(A=T) = P(A=T, B=T, C=T) + P(A=T, B=T, C=F) + P(A=T, B=F, C=F) + P(A=T, B=F, C=T)$$

$$\boxed{P(A=T, B=T) \geq P(A=T, B=T, C=T) + P(A=T, B=T, C=F)}$$

$$
\begin{aligned}
&P(A=F, B=T) \\
&= P(A=F, B=T, C=F) \\
&\quad + P(A=F, B=T, C=T)
\end{aligned}
$$

Bayesian Networks

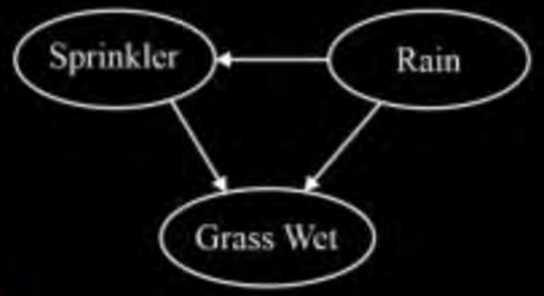
- Let's see one more example
- Find P(W,S,R) and P(W,~S, ~R)

$\Rightarrow P(W, \sim S, \sim R)$

$\Rightarrow P(W|\sim S \sim R) P(\sim S, \sim R)$

$\Rightarrow P(W|\sim S \sim R) P(\sim S|\sim R) P(\sim R)$

$\Rightarrow$ ⓪



Sprinkler

Rain	S ~S
F	0.4 .6
T	0.01 .99

Rain

R ~R
0.2 .8

Grass Wet

Sprinkler		W ~W
~S	~R	0.0 1
~S	R	0.8 .2
S	~R	0.9 .1
S	R	0.99 .01

$P(A, B)$ if A and B are Independent

$\rightarrow P(A)P(B)$

$\Rightarrow P(A=T, B=T, C=T) \Rightarrow \left\{ P\begin{pmatrix} A=T, \\ C=T \end{pmatrix} P\begin{pmatrix} B=T, \\ C=T \end{pmatrix} \right\}$

Independent

A, B are Independent
$P(A,B) = P(A)P(B)$

Bayesian Networks

- Find $P(O, R, \sim H, C)$

$\Rightarrow P(O/R\sim HC) \cdot P(R\sim HC)$

$\Rightarrow P(O/R\sim HC)\, P(R/\sim HC)\, P(\sim H, C)$

$\Rightarrow P(O/R\sim HC)\, P(R/\sim HC)\, P(\sim H)\, P(C)$

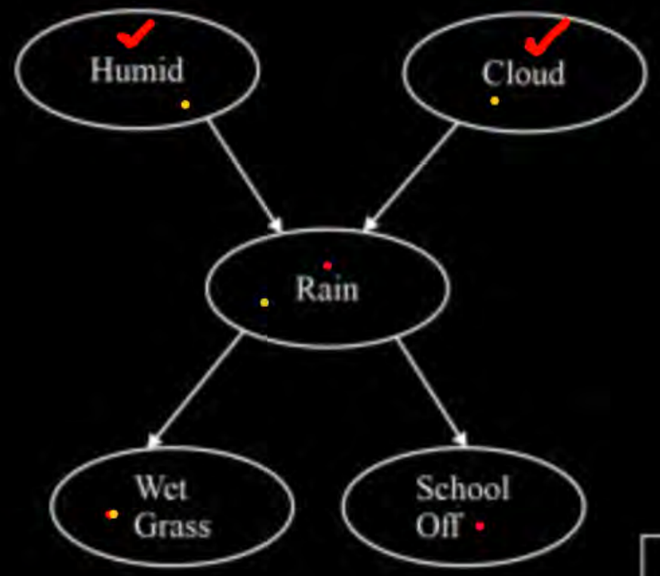
$\Rightarrow .7 \times .7 \times .8 \times .6$

$\Rightarrow .2352$ ✓

P(H)	~H
0.2	.8

H	C	P(R)	~R
H	C	0.9	.1
H	~C	0.4	.6
~H	C	0.7	.3
~H	~C	0.1	.9

R	P(W)	~W
R	0.8	.2
~R	0.1	.9



P(C)	~C
0.6	.4

R	P(O)	~O
R	0.7	.3
~R	0.2	.8

A, B are Independent
$P(A,B) = P(A)P(B)$

Bayesian Networks

- Find $P(O, R, \sim H, C)$

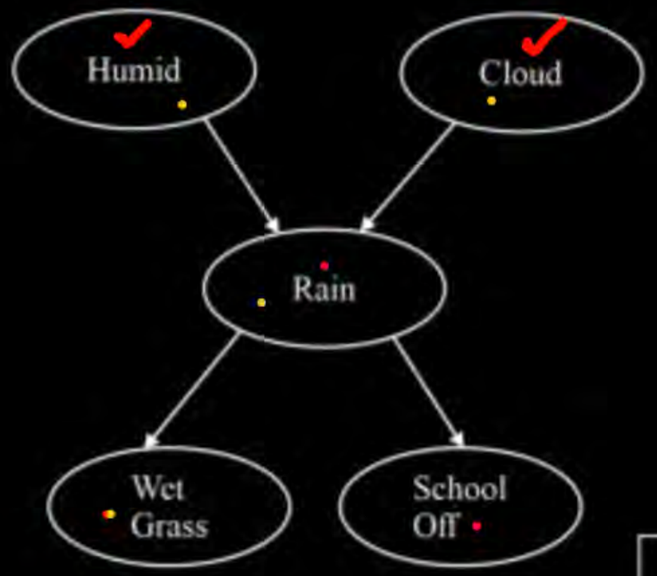
Same → $P(C, \sim H, R, O)$

$P(O/R\sim HC)\,P(R\sim HC)$

lowest variable

P(H)	~H
0.2	.8

H	C	P(R)	~R
H	C	0.9	.1
H	~C	0.4	.6
~H	C	0.7	.3
~H	~C	0.1	.9



P(C)	~C
0.6	.4

R	P(W)	~W
R	0.8	.2
~R	0.1	.9

R	P(O)	~O
R	0.7	.3
~R	0.2	.8

A, B are Independent
$P(A,B) = P(A)P(B)$

Bayesian Networks

- Find P(O,R,~H, C)

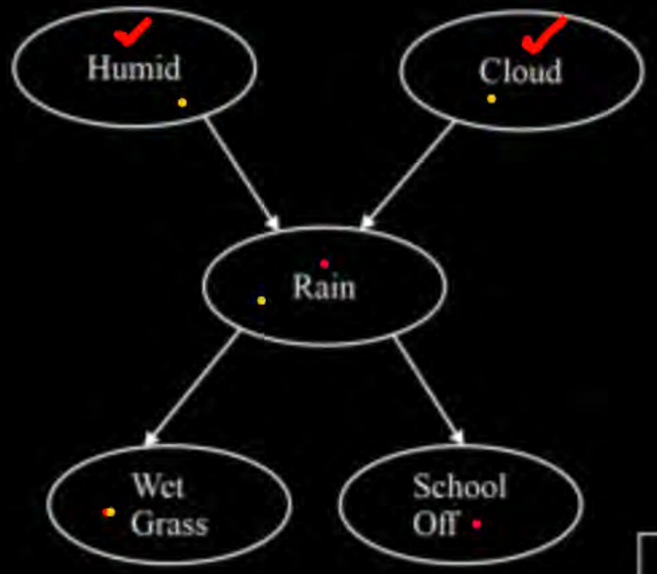
P(WOR)

$\Rightarrow P(W=T, O=T, R=T)$

W and O are Independent to each other

P(H)	~H
0.2	.8

H	C	P(R)	~R
H	C	0.9	.1
H	~C	0.4	.6
~H	C	0.7	.3
~H	~C	0.1	.9



P(C)	~C
0.6	.4

R	P(W)	~W
R	0.8	.2
~R	0.1	.9

R	P(O)	~O
R	0.7	.3
~R	0.2	.8

$$P(R=T) \Rightarrow .9 \times .2 \times .6 + .4 \times .4 \times .2 + .7 \times .8 \times .6 + .1 \times .8 \times .4$$
$$= .508$$

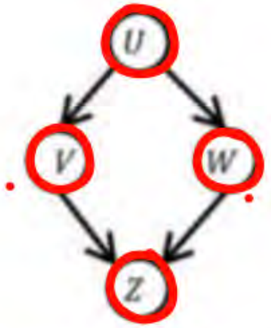
$$\Rightarrow P(W=T, R=T) \cdot P(O=T, R=T)$$

$$\left[P(W=T/R=T) \cdot P(R=T)\right] \left[P(O=T/R=T)\, P(R=T)\right]$$

$$(.8 \times .508)(.7 \times .508)$$

$$= .144$$

Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



	$P(\cdot)$
$U = 0$	0.5
$U = 1$	0.5

	$P(V = 0 \mid \cdot)$	$P(V = 1 \mid \cdot)$
$U = 0$	0.5	0.5
$U = 1$	0.5	0.5

	$P(W = 0 \mid \cdot)$	$P(W = 1 \mid \cdot)$
$U = 0$	1	0
$U = 1$	0	1

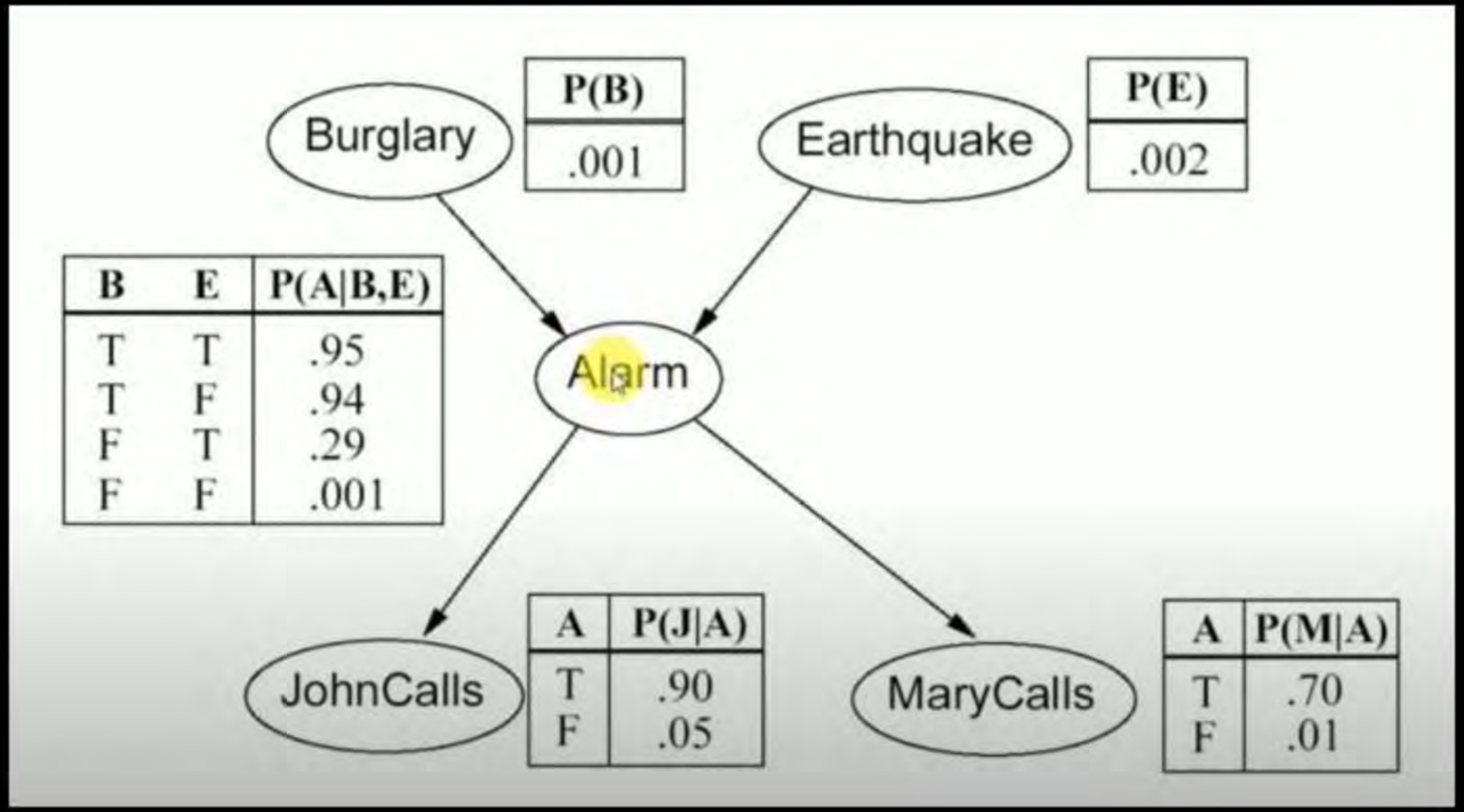
		$P(Z = 0 \mid \cdot)$	$P(Z = 1 \mid \cdot)$
$V = 0$	$W = 0$	0.5	0.5
$V = 0$	$W = 1$	1	0
$V = 1$	$W = 0$	1	0
$V = 1$	$W = 1$	0.5	0.5

The value of $P(U = 1, V = 1, W = 1, Z = 1) =$ _____ (rounded off to **three** decimal places).

$P(Z=1 \mid U=1, V=1, W=1)\, P(U=1, V=1, W=1)$

.5

Bayesian Networks



P(B)
.001

P(E)
.002

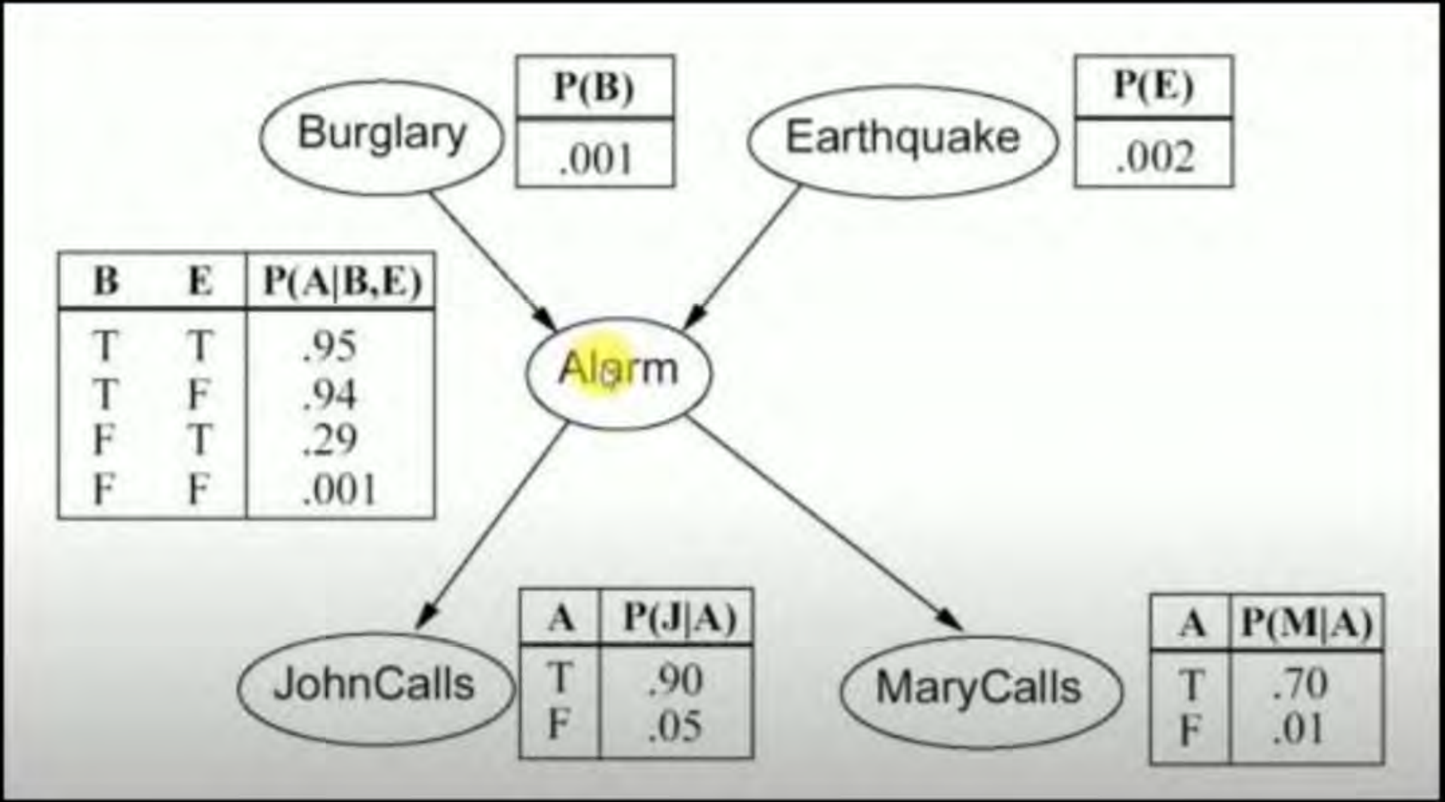
B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

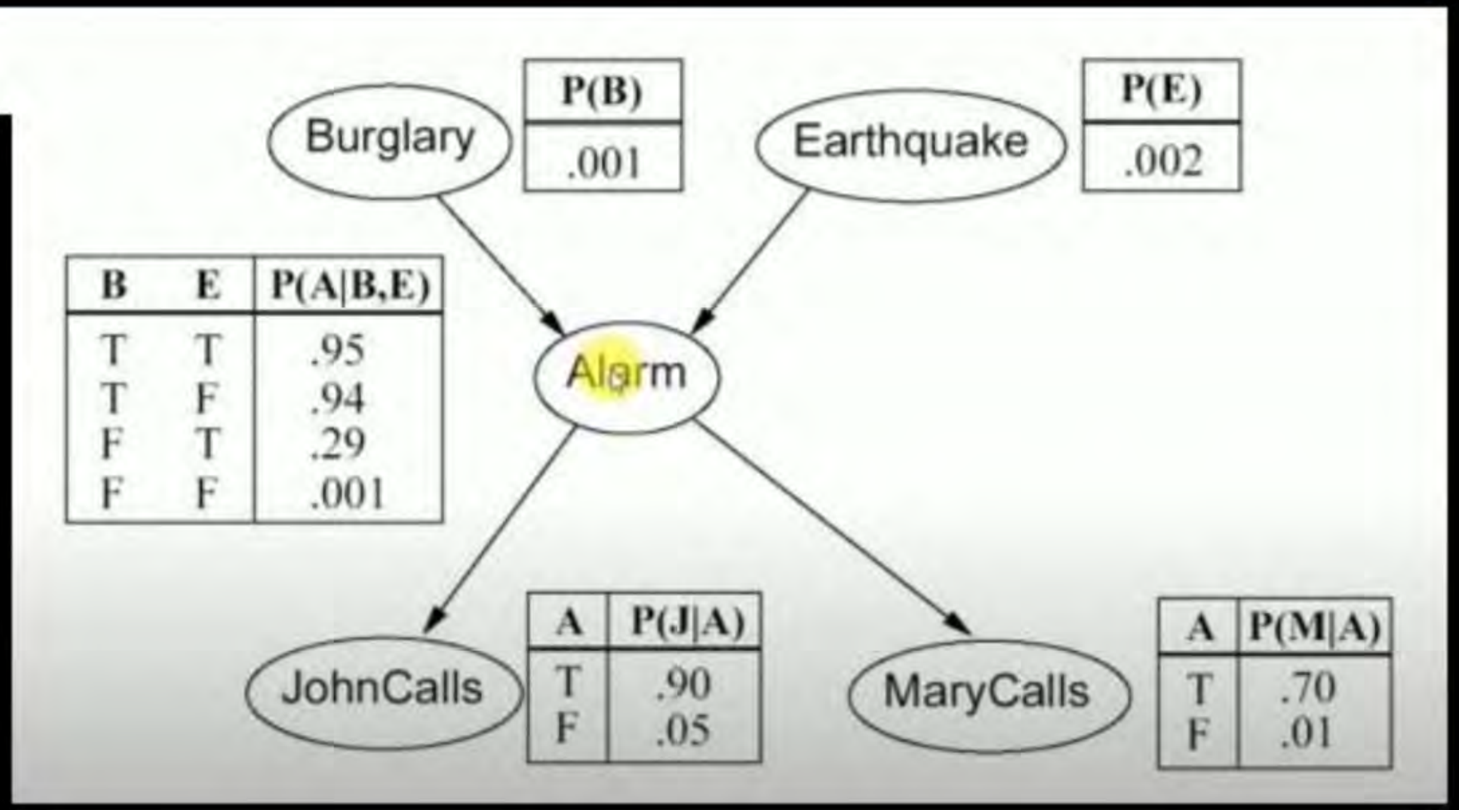
Bayesian Networks

What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?



Bayesian Networks

2. What is the probability that John call?



P(B)
.001

P(E)
.002

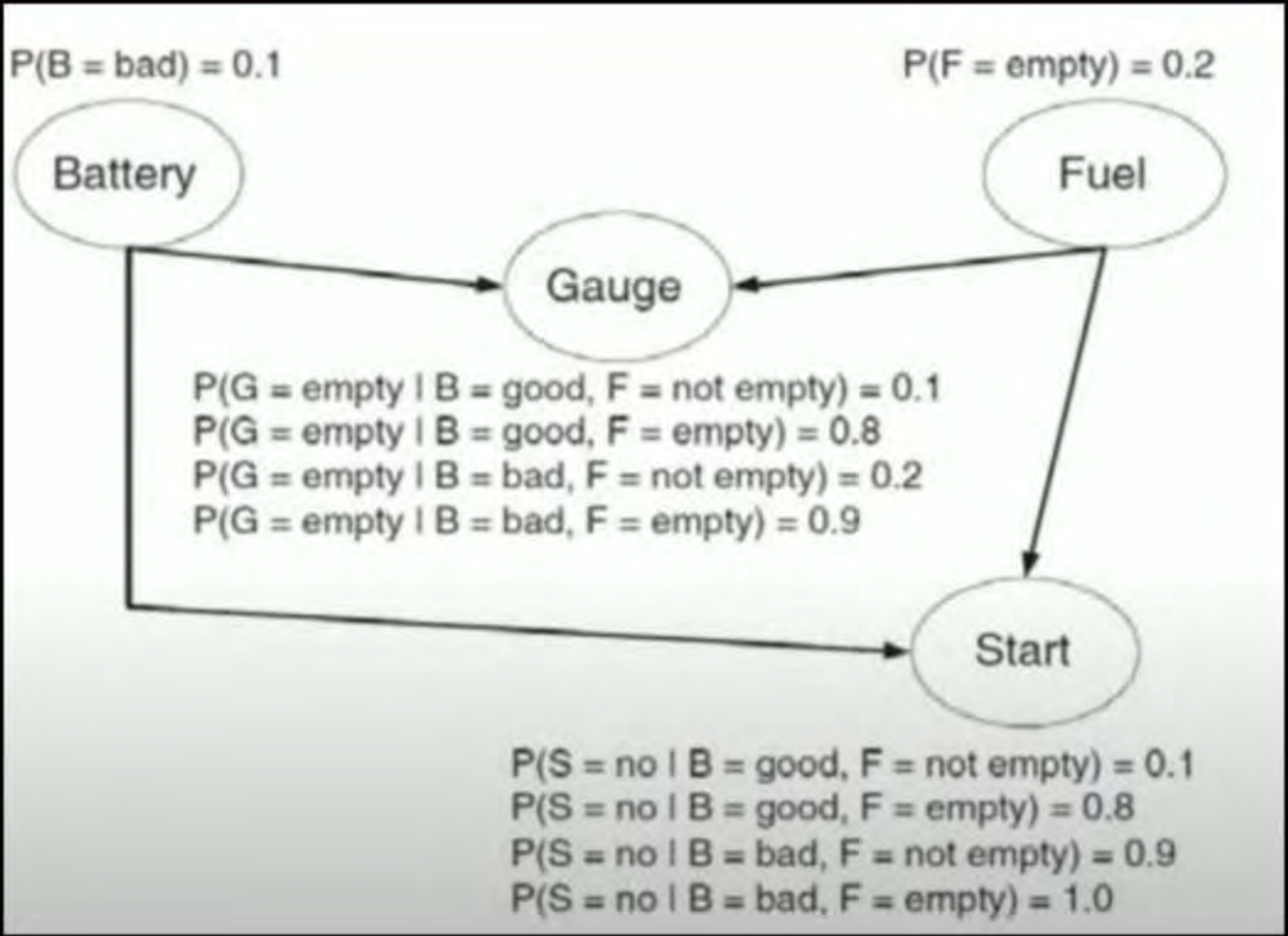
B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

Bayesian Networks

1. P(B=Good, F=Empty, G=Empty, S=Yes)
2. P(B=Bad, F=Empty, G=Not Empty, S=No)
3. Given the battery is bad, Compute the probability that the car will start



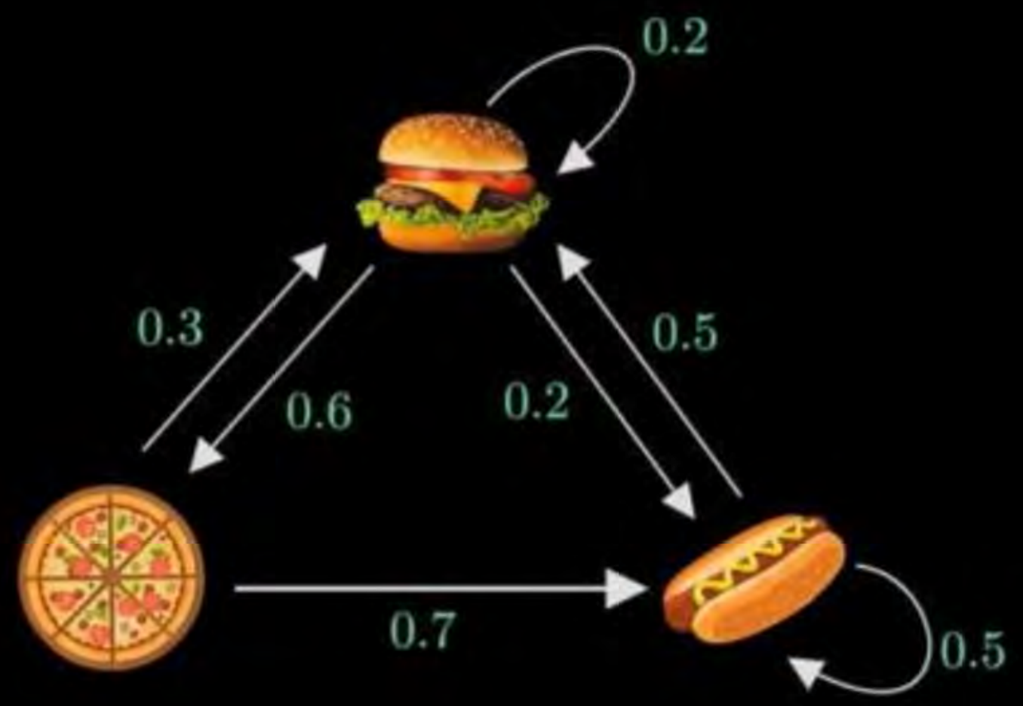
Bayesian Networks

- Components of a Bayesian Network
- Nodes: Each node represents a random variable. Variables can be binary, discrete, or continuous.
- Edges: Directed edges between nodes represent dependencies. If there's an edge from node A to node B, it means that B depends probabilistically on A.
- Conditional Probability Tables (CPTs): For each node, a CPT defines the probability of the node's possible states given the states of its parent nodes.

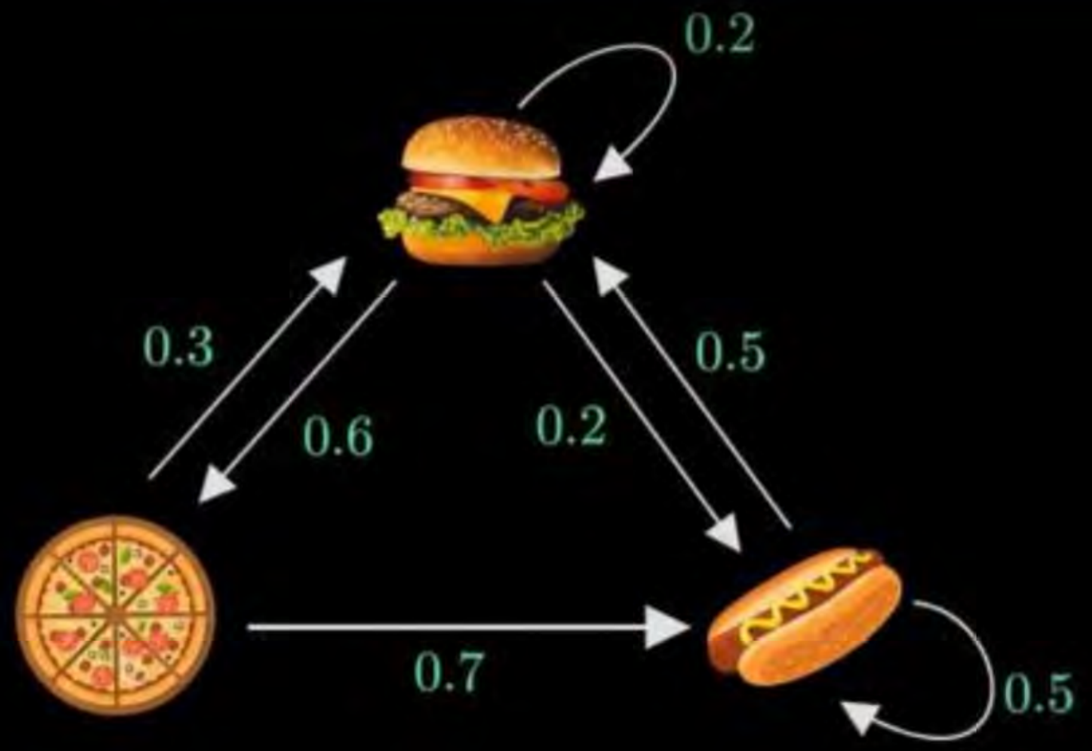
Markov Chain

- A Markov Chain is a mathematical system that undergoes transitions from one state to another within a finite or
- countable number of possible states. It is a type of stochastic process that satisfies the Markov property, which states
- that the future state depends only on the current state and not on the sequence of events that preceded it.

Markov Chain



Probability that hot dog will be served 4th day…



$$P(X_{n+1} = x \mid X_n = x_n)$$

Markov Chain

- Key Concepts: State Space (S):A set of all possible states that a system can be in. For example, in a weather prediction model, the states could be {Sunny, Rainy, Cloudy}.
- Transition Probabilities (P):The probability of moving from one state to another. It is often represented as a matrix P, where P ij represents the probability of moving from state i to state j. The sum of probabilities for transitions from any state must equal 1.

Markov Chain

- Initial State Distribution (π):A probability distribution over the initial states of the system, representing the probability of starting in any given state.
- Transition Diagram:A graphical representation of the states and the transition probabilities between them. Nodes represent states, and directed edges represent the probabilities of moving from one state to another.

Markov Chain

- Example: Weather Prediction Scenario:
- We are modeling the weather using two states: State 1 (Sunny) State 2 (Rainy)Transition Matrix: Suppose we are given the following transition matrix P:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

- $P_{11} = 0.7$: Probability of remaining sunny.
- $P_{12} = 0.3$: Probability of going from sunny to rainy.
- $P_{21} = 0.4$: Probability of going from rainy to sunny.
- $P_{22} = 0.6$: Probability of remaining rainy.

Markov Chain

4. Markov Property

The defining property of a Markov chain is its **memorylessness**, mathematically expressed as:

$$P(X_{n+1} = s_{n+1} | X_n = s_n, X_{n-1} = s_{n-1}, \ldots, X_0 = s_0) = P(X_{n+1} = s_{n+1} | X_n = s_n)$$

This implies that the future state depends only on the present state, not on the sequence of events that preceded it.

Markov Chain

- **Stationary Distribution:**
  A stationary distribution is a probability distribution over the states that does not change as the system evolves. It satisfies the equation:

$$\pi P = \pi$$

  This means that if the system starts in the stationary distribution, it will remain in that distribution after each transition.

THANK - YOU